Micro Set 2. Elasticities — Singapore Hawker Food and Private Transport

Instructions. This booklet contains a single case study on price, income and cross elasticities of demand, followed by six short-answer questions ranging from 2 to 6 marks.

Section Case Study. Hawker Food, Private Transport, and the Singapore Consumer Basket

Table 1.1: Selected Singapore Consumer Demand and Price Indicators, 2019–2024

Indicator	2019	2020	2021	2022	2023	2024
Hawker Centre Meal Price Index (2019=100)	100.0	100.4	102.1	108.6	114.7	117.2
Hawker Centre Meal Demand Index (2019=100)	100.0	71.2	88.4	105.6	109.8	112.4
Restaurant Meal Price Index (2019=100)	100.0	99.8	101.6	107.4	113.0	115.6
Median Household Income, real growth (%)	1.8	−2.2	4.6	−1.0	1.5	2.2
COE Premium, Category A (S$, year average)	32500	36000	58000	82000	87000	92000
COE Category A New Registrations (units)	10120	9580	8200	6900	5100	4200
Bus and Train Ridership (million trips/day)	7.5	4.8	5.4	6.6	7.4	7.7
Private Car Population (thousands)	599	605	612	615	619	624

Source: Adapted from Department of Statistics Singapore, Land Transport Authority, and National Environment Agency.

Extract 1.1: The everyday economics of hawker food and the COE

Singapore's consumer market provides natural experiments in price elasticity. The hawker centre meal — a staple of daily life with over 110 government-run centres housing approximately 6000 stalls — has long been considered the affordable backstop for working-age Singaporeans, with average meal prices around S$4 to S$6. Yet 2022 to 2024 saw the most sustained rise in hawker prices in two decades, with the Hawker Centre Meal Price Index up by 17.2 per cent over the five-year period.

Several factors contributed: cooking-oil and ingredient costs rose sharply during the global commodity shock; manpower costs climbed as foreign-worker quotas tightened; and stallholders faced overheads from longer-term rent adjustments and utilities. "We have absorbed costs for too long. Either we raise prices, or we close," a Bedok hawker told the Straits Times in February 2024. Notably, restaurant meal prices rose by a similar 15.6 per cent over the period, suggesting limited substitution toward restaurants by price-sensitive consumers.

The hawker meal demand pattern is instructive. Demand collapsed in 2020 due to circuit-breaker measures (Demand Index 71.2), then steadily recovered through 2024 (112.4) even as prices rose substantially. The Ministry of Health survey of 2023 found that hawker meals account for approximately 40 per cent of working-Singaporean midday meals, with stronger reliance among lower-income groups. The Public Hygiene Council notes that for many lower-income households, hawker food has few realistic substitutes given its convenience and unit price even after the recent increases.

In private transport, the Certificate of Entitlement (COE) market presents the opposite elasticity profile. The Category A COE premium tripled from S$32500 in 2019 to S$87000 in 2023 and reached S$92000 in 2024. New car registrations in this category fell from 10120 in 2019 to just 4200 in 2024, a 58 per cent reduction. The Land Transport Authority reports that approximately 65 per cent of would-be Category A buyers delayed their purchase, switched to second-hand vehicles, or substituted toward public transport.

Bus and train ridership recovered from a pandemic low of 4.8 million daily trips in 2020 to 7.7 million in 2024 — a 60 per cent rise. The LTA attributes this rise to a combination of price-induced substitution from private transport, returning workplace commuting patterns, and ongoing rail-network expansion. “The COE has performed effectively as a quantity-rationing mechanism, but it is also revealing how price-elastic the demand for private car ownership in Singapore has become,” a transport economist told the Business Times.

Source: Adapted from The Straits Times (February 2024), Land Transport Authority Annual Report 2023, and Ministry of Health Singapore.

Question 1 **[2 marks]**

With reference to Table 1.1, calculate the price elasticity of demand for hawker centre meals between 2021 and 2023, using the simple end-point formula. Show your working.

Question 2 **[2 marks]**

Define “cross elasticity of demand” (XED) and state, with reference to Extract 1.1, whether hawker meals and restaurant meals are substitutes or complements.

Question 3 **[4 marks]**

Using your answer to Question 1 and evidence from Extract 1.1, explain why the demand for hawker centre meals is price-inelastic.

Question 4 **[4 marks]**

With reference to Table 1.1, calculate the price elasticity of demand for Category A COE premiums between 2019 and 2024, and explain what your value implies about the price responsiveness of car buyers.

Question 5 **[6 marks]**

With the aid of two demand-supply diagrams, contrast the effect of a 17 per cent price rise on total expenditure for (a) hawker centre meals and (b) Category A COEs, using the elasticity values you calculated.

Question 6 **[6 marks]**

Using evidence from Extract 1.1, explain how the cross elasticity of demand between private car transport and public transport affects the effectiveness of the COE premium as a tool for managing car population in Singapore.

End of Set 2 — Questions